Figure 1: Navigating to the Digital Forensics toolbox in BackTrack 4

distributions, such as Helix and BackTrack, which are specifically tailored for digital forensics. Since BackTrack is already very popular for its comprehensive tool-kit for security auditing, this article is an attempt to explore the forensics capabilities built into it.

BackTrack, being a Live Linux distribution, is easy to use. One doesn't need permanent installation; any machine can be booted using either a Live DVD or Live USB version of BackTrack. An introductory article, including installation options and general usage, was published in the October 2010 issue of *LFY*. The latest version, BackTrack 4, includes a new boot menu titled 'Start BackTrack Forensics'. This boot menu will start BackTrack in a forensically clean mode, which will not alter the digital data in any manner. This means that the special mode will not automatically mount available drives (thereby avoiding altering the last-mount times of drives). Also, this mode will not use any of the available swap partitions on the hard drive, to avoid altering the data stored in the swap partitions. This can be verified by taking an MD5 snapshot of the drive both before and after using BackTrack in a forensically clean mode, and then comparing both MD5 sums for equality. However, various forensic tools are available in both, the default mode, which is not forensically clean, and the special forensically clean mode just described.

BackTrack Desktop provides easy navigation to various forensics tools by clicking 'BackTrack' -> 'Digital Forensics' on the system menu (Figure 1). A digital forensic investigation generally consists of four major steps (Figure 2):

(i) Data acquisition/data preservation;
(ii) Data recovery;
(iii) Analysis of data for evidence;
(Iv) Reporting of the digital evidence found.

The various forensic tools available under BackTrack are also categorised on the basis of these steps. Listed below are these categories, along with details of some useful tools.

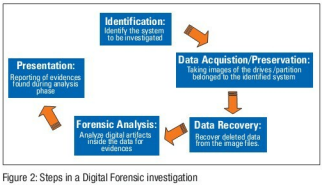

Figure 2: Steps in a Digital Forensic investigation

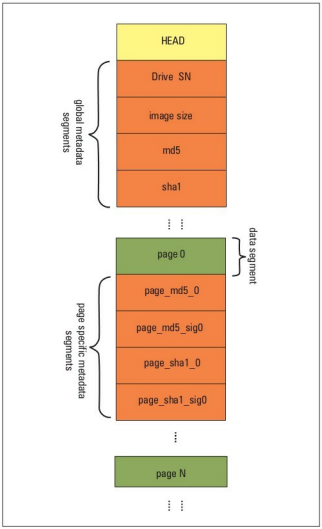

Figure 3: AFF file organisation

Image-acquiring tools

These tools are aimed at taking an exact replica/image (bit-by-bit) of the digital media identified for forensic investigation, without altering the content of the media at all. All methods of forensic analysis aimed at finding digital evidence are then performed on this replica. Acquiring an image of the suspected media helps analyse the media on a high-end workstation, which can fulfil the